



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II

Explainable AI for Finance: Basics + Practitioner Perspectives

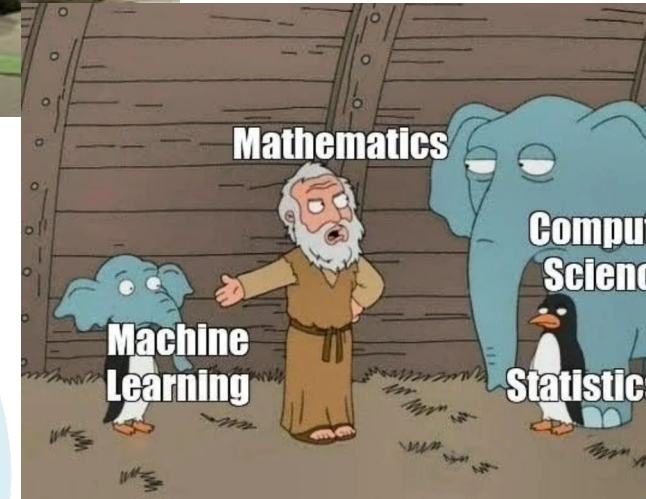
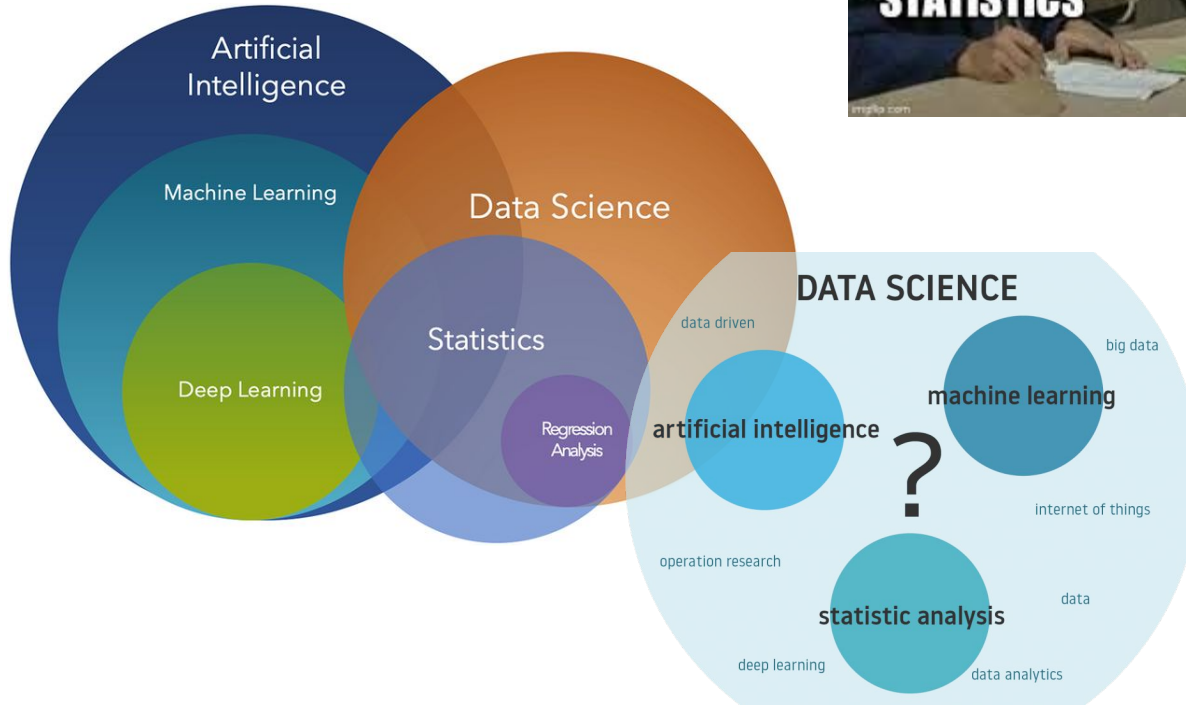
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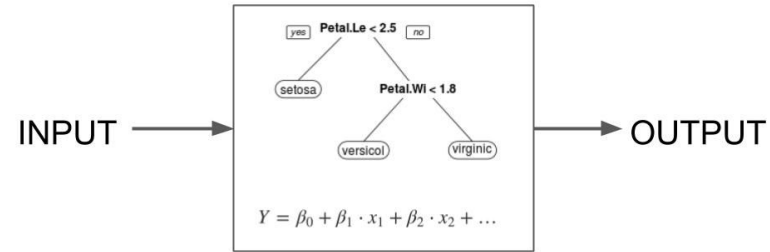
AI and data science

What's the connection?



What are black boxes?

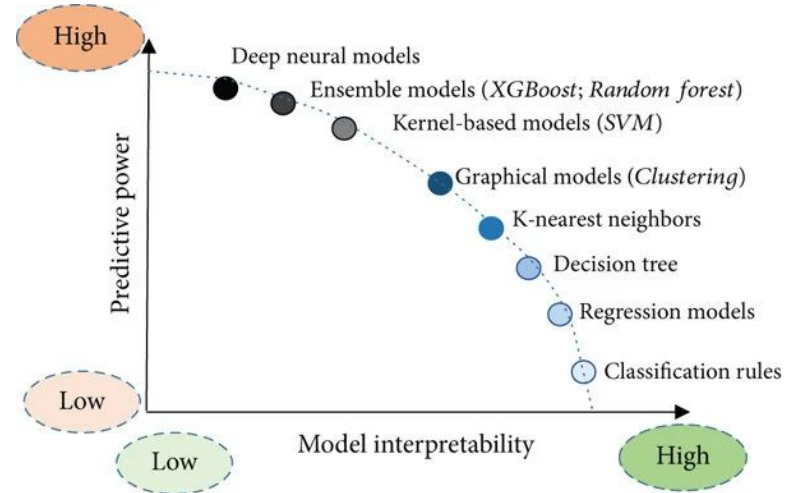
A black box in AI/ML refers to any model whose **internal decision-making process is either unknown to the observer, or known but uninterpretable by humans**. Users can see the input (e.g., a customer's financial data) and the output (e.g., a risk score), but the transformation between them is opaque. This **contrasts with white box models** (e.g., linear regression, decision trees), where every step of the logic can be read and traced.



Accuracy-Transparency Tradeoff

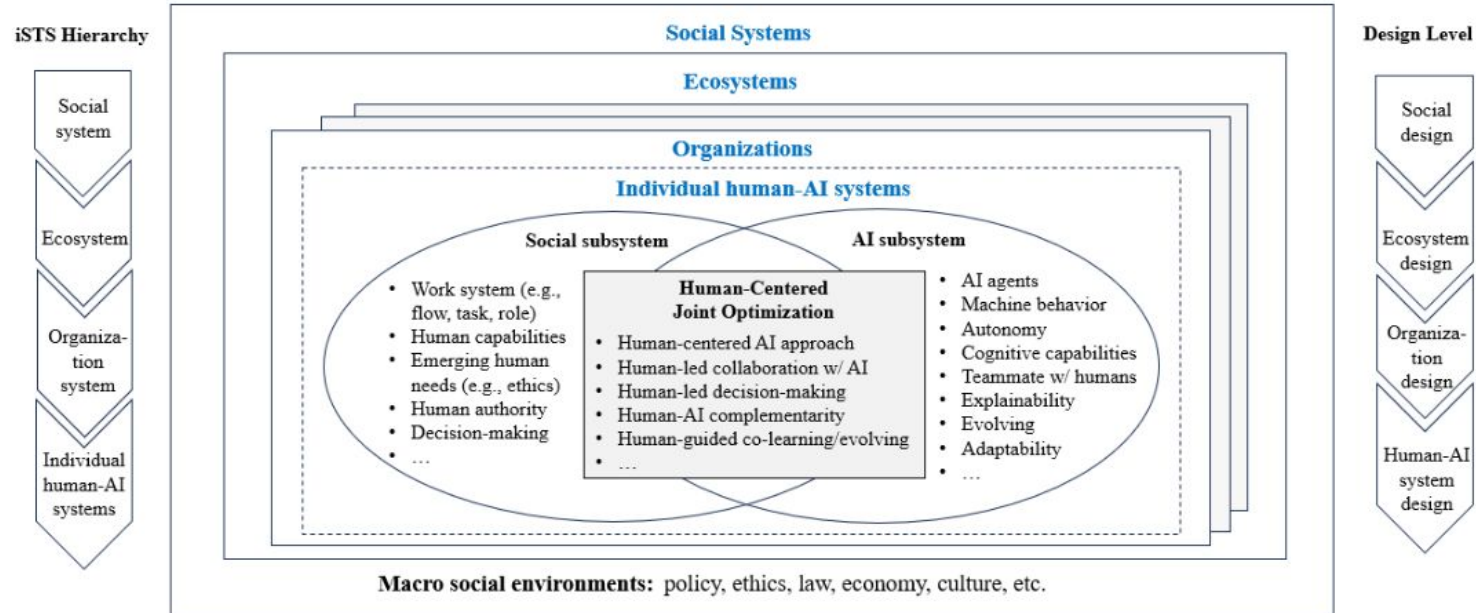
"If I have an extremely performant model, which is a bit harder to explain versus a medium-performing algorithm which is perfectly explainable... you need to balance it off."

— AML FinTech Executive (P10)



Our models sit in a larger system

The AI Act defines AI systems as machine-based systems that can work with autonomy and may learn or adapt after being used. These systems take input data and use it to produce outcomes (predictions, content, recommendations, decisions) **that can influence the physical or digital environments.**



Xu, W., & Gao, Z. (2024). An intelligent sociotechnical systems (iSTS) framework: Enabling a hierarchical human-centered AI (hHCAI) approach. *IEEE transactions on technology and society*, 6(1), 31-46.

But why do we have opacity?

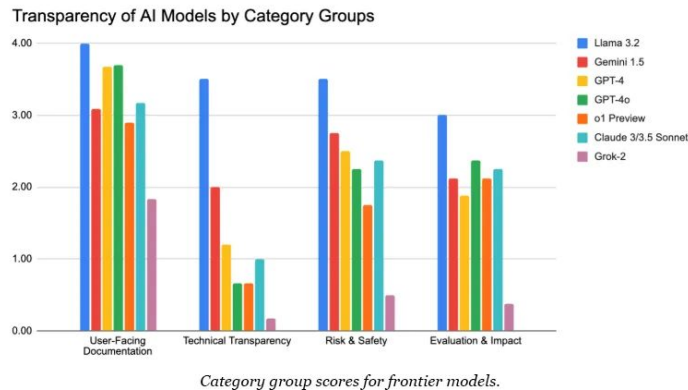
1. Intentional opacity (government or corporate secrecy)

When firms deliberately keep training model secret for business advantage

2. Technical illiteracy

Writing for the computational device demands a special exactness, formality, and completeness that communication via human languages does not.

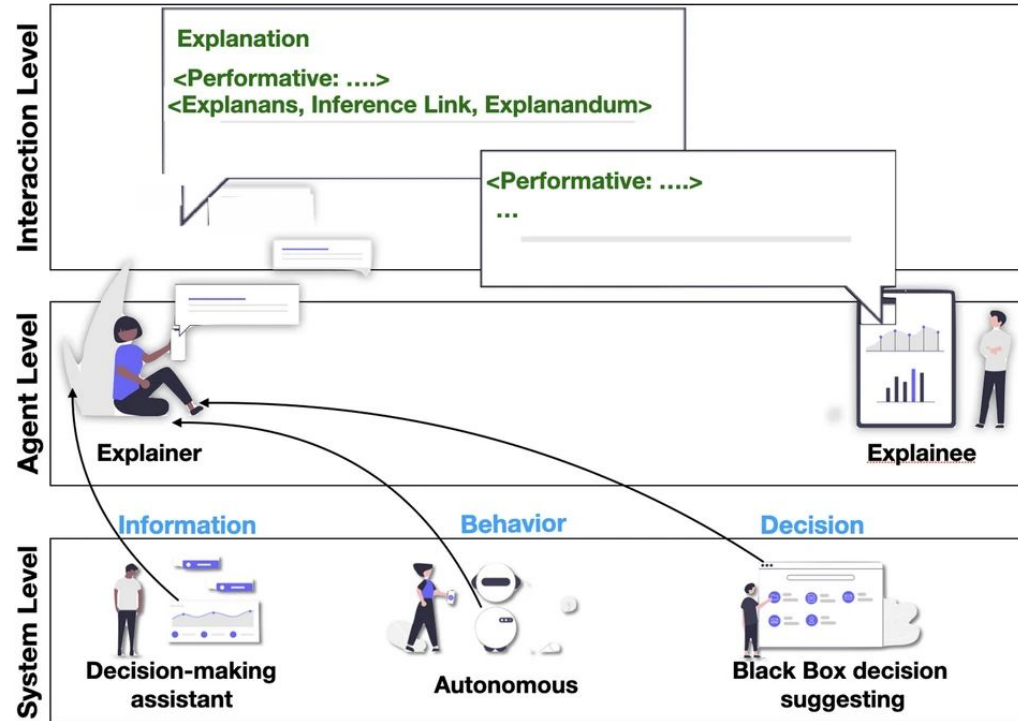
3. Inherent opacity due to complexity and scale



“Even Google’s Gemini 1.5—the most transparent of the closed models—scored just 2 out of 4 on Technical Transparency metrics like model architecture and training data, while the others averaged below 1.”

Source: Americans Responsible for Innovation

XAI models deal with humans



Borrego-Díaz, J., Galán-Páez, J. Explainable Artificial Intelligence in Data Science. *Minds & Machines* **32**, 485–531 (2022). <https://doi.org/10.1007/s11023-022-09603-z>

Explainable AI then implies...

Addressing opacity at the model-level **and beyond**, and **not just technical transparency**.

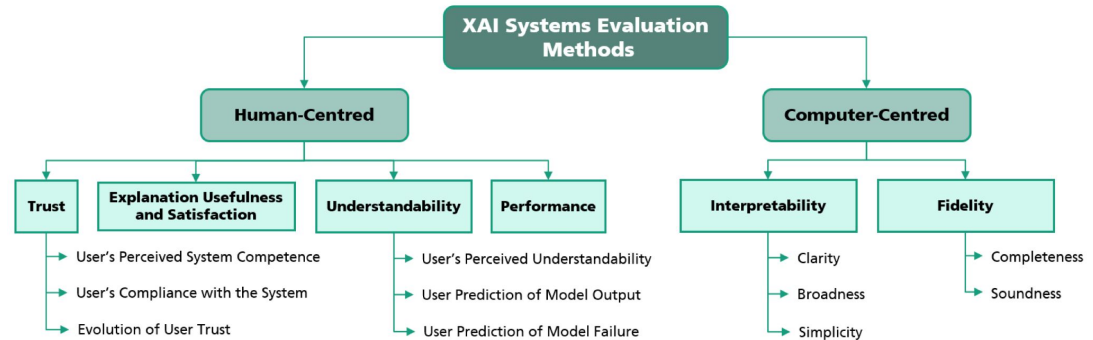
Two strands of XAI (Lopes, 2022):

1. Computer-centric metrics

- a. Fidelity
- b. Robustness

2. Human-centric metrics

- a. Usefulness
- b. Ease of understanding



Lopes, P., Silva, E., Braga, C., Oliveira, T., & Rosado, L. (2022). XAI Systems Evaluation: A Review of Human and Computer-Centred Methods. *Applied Sciences*, 12(19), 9423. <https://doi.org/10.3390/app12199423>

Example of computer-centric XAI for LLMs: Answer relevancy

The Answer Relevancy is defined as the mean cosine similarity of the original question to a number of artificial questions, which were generated (reverse engineered) based on the answer :

$$\text{answer relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(E_{g_i}, E_o)$$

$$\text{answer relevancy} = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|}$$

Where:

- E_{g_i} is the embedding of the generated question i .
- E_o is the embedding of the original question.
- N is the number of generated questions, which is 3 default.

Context precision

Context Precision is a metric that evaluates whether all of the ground-truth relevant items present in the `contexts` are ranked higher or not. Ideally all the relevant chunks must appear at the top ranks. This metric is computed using the `question`, `ground_truth` and the `contexts`, with values ranging between 0 and 1, where higher scores indicate better precision.

$$\text{Context Precision@K} = \frac{\sum_{k=1}^K (\text{Precision@k} \times v_k)}{\text{Total number of relevant items in the top } K \text{ results}}$$

$$\text{Precision@k} = \frac{\text{true positives@k}}{(\text{true positives@k} + \text{false positives@k})}$$

Where K is the total number of chunks in `contexts` and $v_k \in \{0, 1\}$ is the relevance indicator at rank k .

Hint

Question: Where is France and what is it's capital? Ground truth: France is in Western Europe and its capital is Paris.

High context precision: ["France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. Paris, its capital, is famed for its fashion houses, classical art museums including the Louvre and monuments like the Eiffel Tower", "The country is also renowned for its wines and sophisticated cuisine. Lascaux's ancient cave drawings, Lyon's Roman theater and the vast Palace of Versailles attest to its rich history."]

Low context precision: ["The country is also renowned for its wines and sophisticated cuisine. Lascaux's ancient cave drawings, Lyon's Roman theater and", "France, in Western Europe, encompasses medieval cities, alpine villages and Mediterranean beaches. Paris, its capital, is famed for its fashion houses, classical art museums including the Louvre and monuments like the Eiffel Tower",]

XAI for LLMs: broken tool box because originally for ML

What does not work:

- Token probability scores
- SHAP/LIME — no feature importance in token space
- Self-generated justifications — disconnected from actual output scores

What practitioners trust:

- RAG source citations (which document?)
- Visible system prompts (process transparency)
- Behavioral consistency across similar inputs

In **finance** context, why do we need to address opacity? What happens if we do not?

Annex III: High-Risk AI Systems Referred to in Article 6(2)

5. Access to and enjoyment of essential private services and essential public services and benefits:

- (b) AI systems intended to be used to **evaluate the creditworthiness of natural persons or establish their credit score**, with the exception of AI systems used for the purpose of detecting financial fraud;
- (c) AI systems intended to be used for risk assessment and pricing in relation to natural persons in the case of **life and health insurance**;

So what?

DIVE BRIEF

Apple Card under investigation for alleged gender discrimination

Published Nov. 11, 2019



Anna Hrushka

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Vol. 8 No. 1 (2025)

Double discrimination: Algorithmic amplification of gender bias in African fintech credit scoring—a 10-algorithm audit reveals 37% underfunding penalty against women-led SMEs

ARTICLES

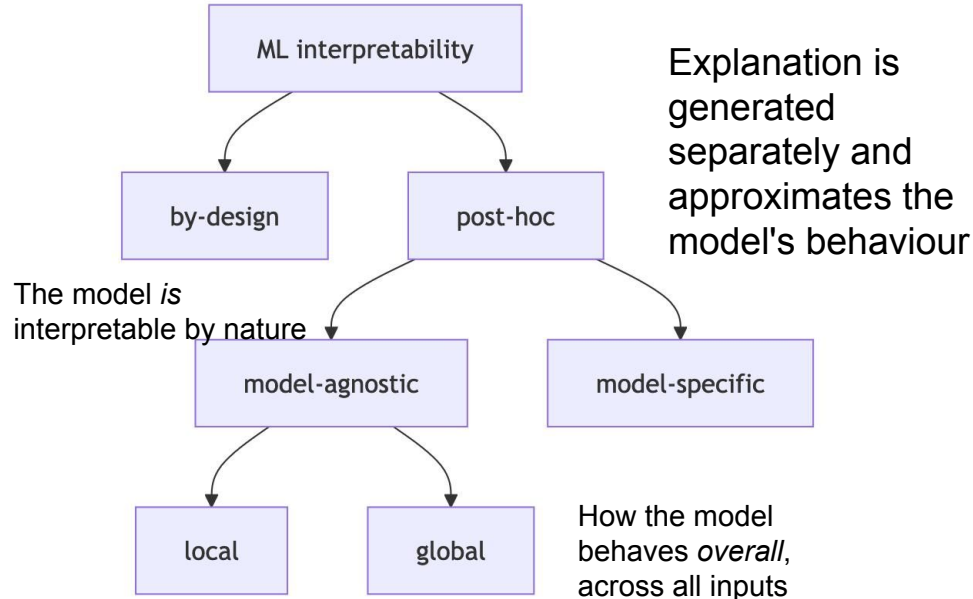
<https://doi.org/10.71350/3062192576>

Published 2025-07-14

Simon Suwanzy Dzreke* , Semefa Elikplim Dzreke*

communities. According to a [2022 UC Berkeley study](#) on fintech lending, averaging across the distribution of these products in the U.S., African American and Latinx borrowers are charged nearly **5 basis points in higher interest rates** than their credit-equivalent white counterparts—**amounting to \$450 million in extra interest per year.**

Explainable AI



Why this *specific* prediction was made for this *specific* instance

Molnar, C. (2025). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (3rd ed.).

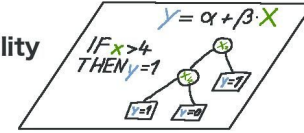
christophm.github.io/interpretable-ml-book/

Humans



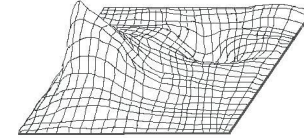
↑ inform

Interpretability Methods



↑ extract

Black Box Model



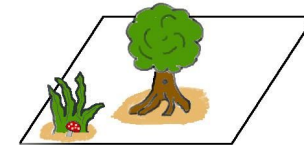
↑ learn

Data

	K	X	K	.	.	.	.	K
1	1	5	4	0	.	.	.	0
2	1	2	0	.	.	.	.	0
3	1	5	4	0	.	.	.	0

↑ capture

World



Examples of XAI in finance (CFI)

Technique	How It Improves Explainability	Example Use in Finance
SHAP (Shapley Additive Explanations)	Identifies which features influenced a model's decision and by how much.	Credit risk models showing how income, credit utilization, and payment history impact loan approvals.
Counterfactual Explanations	Explains what changes would have led to a different decision.	Mortgage applications: "If your debt-to-income ratio was below 43%, your loan would be approved."
Interpretable Models	Uses inherently explainable structures like decision trees and regression models.	Simple credit scoring models where decision pathways can be understood at a glance.
Rule Extraction	Converts black-box model decisions into readable guidelines.	AI-driven investment recommendations translated into human-readable strategies.

Issues with extant XAI models

Open Issue	Core Problem	Most Affected Financial Domain
No definition of explainability	Cannot measure or enforce it	All domains
No evaluation metrics for XAI quality	Cannot compare or validate XAI tools	Credit, fraud, trading
Accuracy–interpretability trade-off	Best models remain opaque	Credit scoring, AML
Audience-dependent explanations	Technical XAI useless for decision-makers	Credit, compliance, retail
Faithfulness of post-hoc methods	SHAP/LIME may misrepresent the model	All post-hoc applications
Regulatory fragmentation	No enforceable global standards	All regulated domains

Černevičienė, J., Kabašinskas, A. Explainable artificial intelligence (XAI) in finance: a systematic literature review. *Artif Intell Rev* **57**, 216 (2024).

<https://doi.org/10.1007/s10462-024-10854-8>

SHAP is theoretically good to implement but not compliant with human-centric goals

Kumar, I. E., Venkatasubramanian, S., Scheidegger, C., & Friedler, S. (2020). Problems with Shapley-value-based explanations as feature importance measures. *Proceedings of the 37th International Conference on Machine Learning*, 119, 5491–5500. Proceedings of Machine Learning Research.

Problems with Shapley-value-based explanations as feature importance measures

I. Elizabeth Kumar¹ Suresh Venkatasubramanian¹ Carlos Scheidegger² Sorelle A. Friedler³

Abstract

Game-theoretic formulations of feature importance have become popular as a way to “explain” machine learning models. These methods define a cooperative game between the features of a model and distribute influence among these input elements using some form of the game’s unique Shapley values. Justification for these methods rests on two pillars: their desirable mathematical properties, and their applicability to specific motivations for explanations. We show that mathematical problems arise when Shapley values are used for feature importance, and that the solutions to mitigate these necessarily induce further complexity, such as the need for causal reasoning. We also draw on additional literature to argue that Shapley values are not a natural solution to the human-centric goals of explainability.

1. Introduction

Machine learning models are increasingly being used to replace human decision-making for tasks involving some kind of prediction. As state-of-the-art predictive machine learning models become increasingly inscrutable, there has been an increase in concern that the black-box nature of

by the model, and the game is the prediction of the model. A variety of methods to assign feature influence using the Shapley value have recently been developed (Lipovetsky & Conklin, 2001; Štrumbelj & Kononenko, 2014; Lundberg et al., 2018; Datta et al., 2016b; Merrick & Taly, 2019; Frye et al., 2019; Aas et al., 2019).

In this paper, we demonstrate that Shapley-value-based explanations for feature importance fail to serve their desired purpose in general. We make this argument in two parts. Firstly, we show that applying the Shapley value to the problem of feature importance introduces mathematically formalizable properties which may not align with what we would expect from an explanation. Secondly, taking a human-centric perspective, we evaluate Shapley-value-based explanations through established frameworks of what people expect from explanations, and find them wanting. We find that the game theoretic problem formulation of Shapley-value-based explanations do not match the proposed use cases for its solution, and thus caution against their usage except in narrowly constrained settings where they admit a clear interpretation.

We describe the different Shapley-value-based explanation frameworks in Section 2, and present our two-part critique in Sections 3 and 4. We discuss these results and provide some suggestions in Section 5.

What regulators say: it's important. But finance industry does not have capability. And it is not intuitive nor help us execute GDPR.

- **Hard to understand:** Current XAI outputs (plots, attributions, counterfactuals) are too technical for ordinary people and many non-technical professionals.
- **Legally incomplete:** No single method clearly satisfies GDPR requirements for a “meaningful explanation” and a real right to contest decisions.
- **Controller vs subject tension:** Explanations often serve the bank’s perspective (compliance, IP protection) more than the customer’s need to understand and challenge.
- **Form matters as much as method:** Experts want plain-language narratives, realistic “what-if” scenarios, and evidence of consistent treatment, not just feature-importance plots.



State, L., Bringas Colmenarejo, A., Beretta, A., Ruggieri, S., Turini, F., & Law, S. (2025). *The explanation dialogues: An expert focus study to understand requirements towards explanations within the GDPR*. Artificial Intelligence and Law. Advance online publication. <https://doi.org/10.48550/arXiv.2501.05325>

Let's try SHAP! How toxic are you?

<https://poloclub.github.io/webshap/>



If our XAI models are not intuitive, can data science help?

Yes. Over 90% of the surveyed general audience finds the narratives generated by SHAPstories convincing. (Martens et al., 2024)

Bottomline: The same tool that satisfies a data scientist completely fails a compliance officer. Explainability is audience-dependent.

<https://www.storytellingwithdata.com/>

BEFORE: SHOWING DATA

July Results



ANALYSIS FINDINGS: Total unit decline in 12-month (-26%) and 48-month (-36%) from August to July, while cash grew (10%).

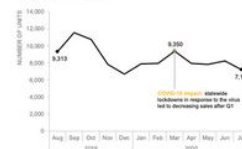
AFTER: STORYTELLING WITH DATA

Consumer behavior has shifted

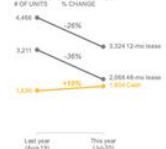
Total units sold is down 22.7% YoY and down 23% since the impact of COVID lockdowns in the US. Purchase type composition has evolved over time. How might this impact our go-forward strategy?

The COVID impact caused a sharp decrease in the number of units sold as of July. We expect this decline to level out by each year-end due to discussions with key customers about resuming their purchasing cycles. As a result of the market and access to financing, the 12 and 48 month leases are decreasing in utilization. Cash purchases have increased 10% in the last 12 months. We expect this to continue.

Total sales over time



Sales by purchase type



Footnote here with data source, relevant metric definitions, assumptions, methodology, etc.

Open research questions

- Can we close the explainability gap with better feature engineering?
- Better to improve models or just the presentation?
- How do you measure user understanding of an agentic system?

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What the New Cambridge Global Report Tells Us About Explainability Gap

The picture that emerges is not one of crisis, but of a sector in mid-transition: adoption is high, value is emerging, and yet protection & bias explainability and governance still have some catching up to do.

[READ THE LATEST](#)

Lack of explainability

37% (n=126) 19pp 54% (n=71)

Governance is control gap

33% (n=113) 17pp 50% (n=62)

Lack of understanding

28% (n=94) 15pp 43% (n=54)

Culture & operational resilience

46% (n=156) 13pp 59% (n=74)

Loss of human oversight

27% (n=94) 14pp 41% (n=51)

Data integrity & model drift

42% (n=52) 13pp 55% (n=187)

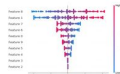
42% (n=142) 12pp 54% (n=68)

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An Introduction to SHAP for Finance

This article is part of the "Introduction to XAI methods in Finance" series.

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